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Product Design and Manufacturing

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Zero-Screw Car Table

Project Report

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Abstract

Product design is the process of conceptualizing, developing, and iterating products that answer users' issues or meet specific market needs. Understanding the end-user customer, or the person for whom the product is being built, is critical to a successful product design. The project's main aim is to develop a portable table that will succeed in the market, keeping in mind all the following steps starting with opportunity identification and going through determining customer needs, setting specifications, performing a patent search, concept generating, concept selection, and Concept testing. The result of each step was the input for the following steps. Other topics such as design for manufacturing, managing projects, and product development economics. The team's biggest challenge in the project was the lack of technical information in specific topics and the lack of expertise.

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1.0 Introduction

Product design and manufacturing is an enormous field in every single business, from automobiles and airplanes to even electronics. The field starts by identifying opportunities to pursue and goes up to delivering the final product to customers. According to the authors from the car connection website [1], people who own cars spend an average of 8 hours a week behind the wheels, and according to the survey done by the team, 95.2% of all participants eat at least once a week inside their cars. Surprisingly all participants who ate in the car faced problems while eating.

Cars were not designed for eating; however, if a product can be designed in which most of the customer needs are satisfied, then cars will be a great place to enjoy food. Restaurants nowadays give more attention to takeaway orders instead of dine-in. In addition, people who are required to do work in the car don't feel comfortable. As a solution to the above problem, the team discussed designing a portable car table that can be easily fixed and removed. The table is fixed in the place of the cupholders and then spread throughout the car's width. The table can also be used for outdoor use by fixing four legs at each corner. The main idea is to be able to eat in the car without facing difficulties like where to put the food and drinks. Also, it aims to make the idea of studying and finishing work in your car much easier.

This product targets an extensive range of people (adults) and everyone who goes on hikes or any outdoor activity. Moreover, students are another target group as they can utilize their free time waiting in the car to study and finish any assigned tasks. People interested in outdoor activities can benefit from the table as it does not need any tools or screws, it can be assembled easily.

2.0 Opportunity identification

An opportunity is a form of a concept or an idea of a new product. In this stage, the team started collecting different opportunities that have good chances to succeed in the market and gain customer satisfaction. The following 3 opportunities were gathered:

2.1 Portable Furniture

Students face problems when they need to move to another apartment for example the cost of renting a truck to move their furniture and it is very difficult to move it to the upper floors due to the heavyweight. Moreover, the people who like to have outdoor parties face problems with providing suitable and modern furniture, so they need to rent them from specialized companies. The Furniture can be assembled using joints. The main idea is to be able to take it apart again to be able to use them many times with easy steps of assembly and disassembly. The furniture can help the users to maintain the spaces and make the palaces look modern and clean. Also, it can be used whenever the user needs it. Figure 1 and 2 represents an example of the opportunity.

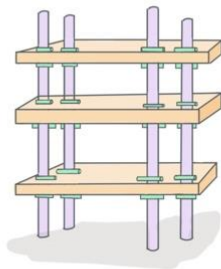


Figure 1: Example 1 of portable furniture

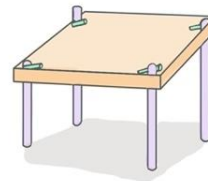


Figure 2: Example 2 of portable furniture

2.2 Whiteboard eraser

One of the main issues that teachers and doctors face during teaching is erasing the whiteboard. It's both exhausting and time-consuming. This problem could be easily encountered by the product that our team has come with. A long vertical rod with an erasing foam on one side is attached to the whiteboard both from up and down. It can slide horizontally and erase all the whiteboard in a matter of seconds as shown in figure 3.

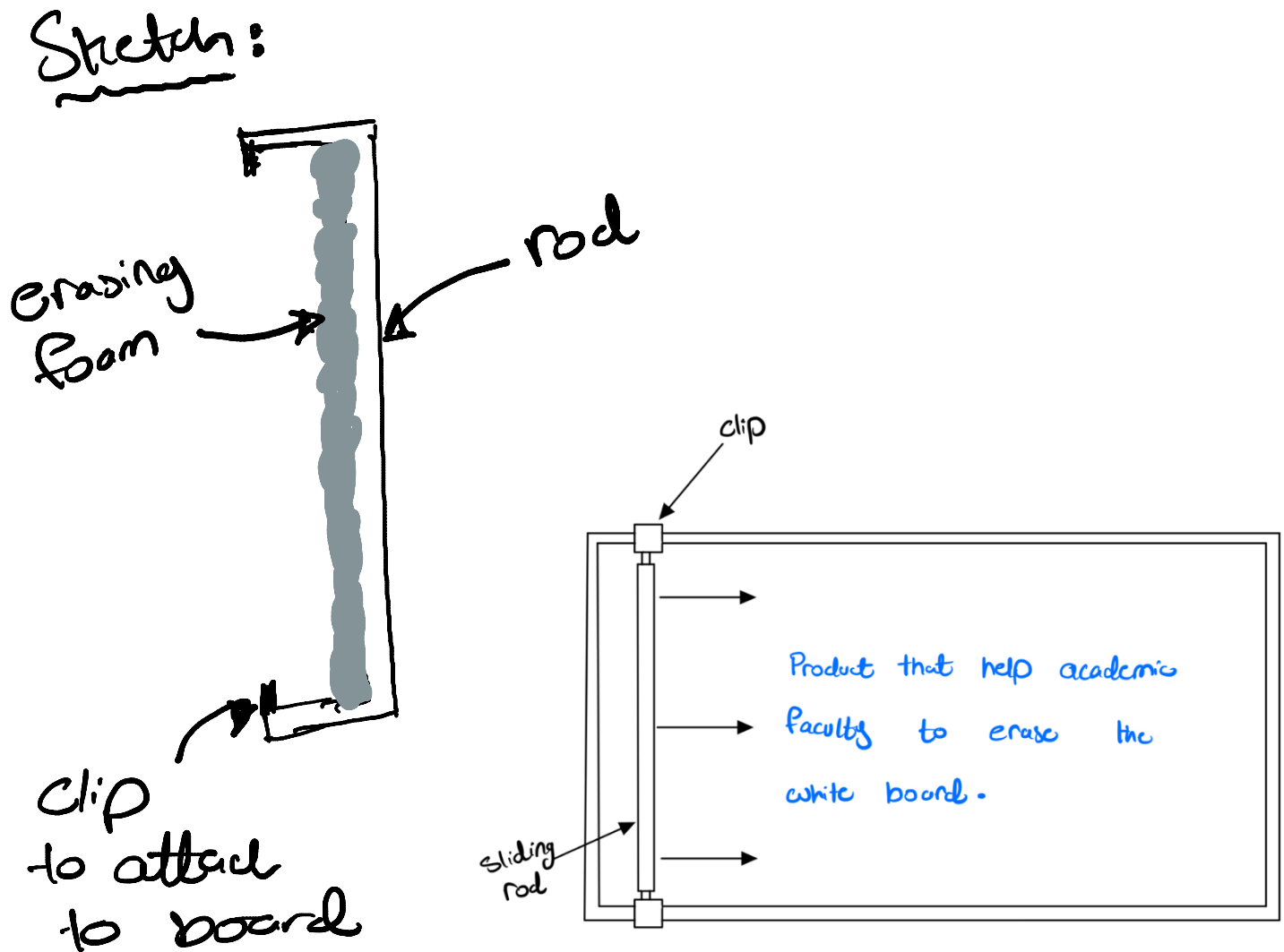


Figure 3: Concept of second opportunity

2.3 Portable Car table

The team decided to hold a meeting outdoors to discuss what opportunity to pursue. The team ordered food takeaway and headed to where the meeting was planned to be. However, the humidity was high that night, so the meeting was held in the car. During eating in the car, a sauce from a burger fell into a dishdasha and into the car leading to a mess. This is where one of the team members asked, “why don’t we design a portable table?”. An example is shown in figure 4.

The team discussed all opportunities gathered and decided to go with opportunity 3 because they all agreed that it has a higher chance of success.



Figure 4: Car Table example

Then after deciding what opportunity to go with. A meeting was held, and the mission statement was constructed as shown in table 1.

Table 1: Mission Statement

Mission statement: Portable Car Table

Product Description	• A Portable Foldable car table with a picnic function.
Benefit proposition	• Ease in dining in the car. • 2 in 1 table. (Dinning in the car and outside) • Work comfortably in the car
Primary Market	• Every Car owner who wishes to dine with ease in the car.
Secondary Market	• Adventurers, hikers, and picnic lovers.
Assumptions and Constraints	• Easy to install and remove. • Good surface texture. • Lightweight and small size.
Stakeholders	• User • Production • Management • Salesforce.

3.0) Customer Needs

Gathering customer needs is one of the most essential part of product design. The marketing team is responsible for conducting interviews, focus groups, and surveys. The team decided to conduct 5 interviews, a sample of the interviews is shown in figures 5,6 and one survey, figure 7 represents a sample of the questions asked in the survey. Then after contacting the customers and gathering data from them, the data was handed to the design team. From the interviews and the 65 responses gathered from the survey, the needs were then converted to need statements using the 5 laws learned in the course. After that, the level of importance for the needs gathered was considered and the results are shown in Table 2.



Figure 5: Interview 1



Figure 6: Interview 2

How often do you eat in the car?
Do you face a problem while eating food in the car?
How likely would you like having a new product that helps you to eat in the car?
How likely would you like a new product that helps you working in the car?

Figure 7: Survey Questions

Table 2: Needs of customers and need statements

	Needs of customers	Need statements	Imp.
1	Flat surface to eat on	The table has a flat surface.	2
2	Eat and drive at the same time	The table is stable while driving.	2
3	Can also be used for makeup	The table can be used for holding makeup.	5
4	Useful and take quite a little place in the car	The table is small	3
5	Unique	The table is unique	3
6	Foldable table	The table is foldable	2
7	Good product (quality)	The table is made of good material	1
8	Can put it easily	The table is easy to fix /install	1
9	Easy to clean	The table can be cleaned easily	2
10	Removable	The table can be removed easily	2
11	Comfortable	The table is comfortable	2
12	Don't distract driver vision	The table is not on the face of the driver.	2
13	Portable charger	The table has a portable charger	5
14	Cupholder	The table contains a cup holder	3
15	Easy carry	The table is portable	1
16	Last for a long time	The table is durable	2
17	No sharp corners	The table is safe to use	2
18	Large surface area	The table surface area is large.	1
19	Not slippery	The food grips on the table surface	1
20	Comes with a place for forks and spoons	The table has a storage area	4
21	Accessibility for everyone	The table is cheap	3

4.0 Product Specification

After getting the need statements, setting the ideal and marginal specifications is an essential step to knowing what target specifications the team should pursue. The team looked for competitive

products in the market to benchmark, however, they couldn't find any product. Based on a few searches in google, the target specifications were constructed as shown in table 3.

Table 3: Target Specifications

	Metric	Unit	Ideal value	Marginal value	Imp.
1	the angle of the table surface	Degrees	90°	$89^{\circ} \leq x \leq 91^{\circ}$	2
2	vibration on the table surface	Hz	20 Hz	$x \leq 30 \text{ hz}$	2
3	circle volume	cm cube	-	-	5
4	The volume of the table	cm cube	3200 cm^2	$x \leq 3200 \text{ cm}^2$	3
5	Color	cm cube and list	None	none	3
6	number of folds	Folds	2	4	2
7	product life	Years	3 years	$x \geq 3 \text{ years}$	1
8	time to install	Seconds	30 sec	$x \leq 60 \text{ seconds}$	1
9	time to clean	Seconds	5 sec	$x \leq 5 \text{ seconds}$	2
10	time to remove	Seconds	20 sec	$x \leq 30 \text{ seconds}$	2
11	special tool requires	List	-	-	2
12	Block view or not	List	-	-	2
13	special tools require	List	-	-	5
14	special tools require	List	-	-	3
15	the total mass	kg	0.8 kg	$x \leq 2 \text{ kg}$	1
16	Durable or not	list	-	-	2
17	special tool requires	list	-	-	2
18	surface area	cm ²	1600 cm^2	$x \leq 1600 \text{ cm}^2$	1
19	friction coefficient	u	0.8	$0.5 \leq x \leq 0.8$	1
20	special tool requires	list	-	-	4
21	unit manufacturing price	OMR	7 OMR	$x \leq 10 \text{ OMR}$	3

1.

Publication Date: 04/29/1975

Brief description: the invention includes a panel, such as plastic, that is connected to one pair of fixed-length legs with sufficient lateral spacing to engage the floor on either side of the hump and the second pair of legs with less spacing, these legs being adjustable in length to allow the customers to level the table. All the legs are made of properly bent steel rods as shown in figure 8.

2.

Publication Date: 11/16/1982

Figure 9: Patent 2

3.

Name: Detachable car seat tables

Publication Date: 06/25/1985

Patent number: United States Patent 4524701

Brief description: A removable car seat table that can be temporarily fastened to the carpet on an automobile's floor, particularly the carpet covering the hump containing the drive system on the floor. The table includes a tray section with the back portion resting on the upper surface of a car seat and the forward portion supported by a vertical support member secured between said tray section and a lower connection section with a horizontal plate having to oppose laterally adjustable side wings with inwardly projecting slanted teeth on said side wings as shown in figure 10.

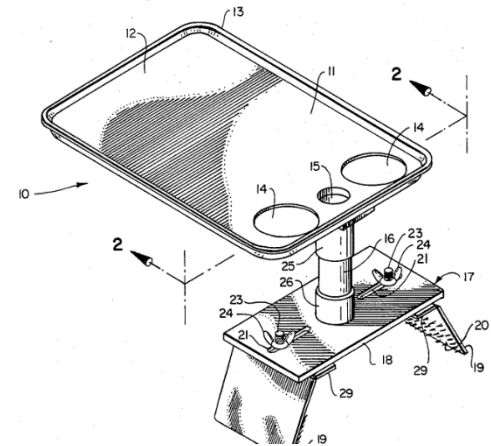


Figure 10: Patent 3

4.

Name: Adjustable car table

Publication Date: 12/15/1992

Patent number: United States Patent 5170720

Brief description: A removable car seat table that can be temporarily fastened to the carpet on an automobile's floor, particularly the carpet covering the hump containing the drive system on the floor. The table includes a tray section with the back portion resting on the upper surface of a car seat and the forward portion supported by a vertical support member secured between said tray section and a lower connection section with a horizontal plate having to oppose laterally adjustable side wings with inwardly projecting slanted teeth on said side wings as shown in figure 11.

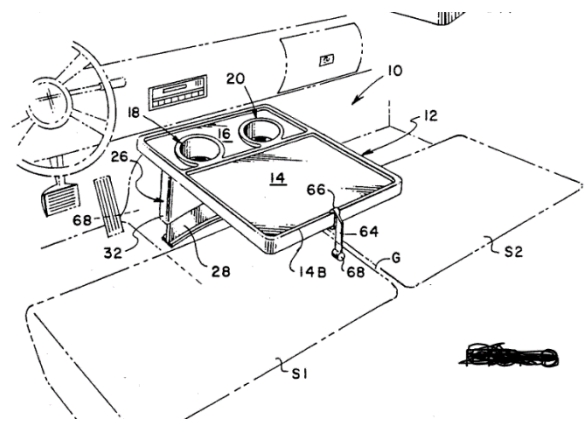


Figure 11: Patent 4

6.0 Concept Generation

In this stage, an external and internal search was used to generate concepts. For external search, a patent search was used, and the result of the search is shown in section 5. As for internal search, the team used brainstorming and the gallery method in the many meetings held. In the end, 5 concepts were generated as shown below.

Concept A)

The table has a 0.1 m² surface area and is composed of lightweight but durable plastic. The scratch-resistant plastic also has a high coefficient of friction, ensuring that the meal or snack does not slip. The two clips depicted in figure 12 are used to secure it to the steering wheel. It's simple to assemble and disassemble, and it comes with a cup holder that fits standard-size cups.

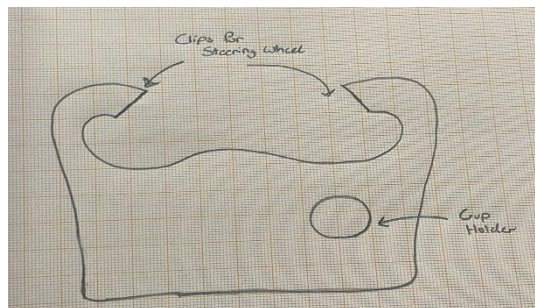


Figure 12: Concept A

Concept B)

The table has a 0.9m² surface area and is made of wood, which appears elegant and luxurious. The weight is estimated to be 0.5 kg. By sucking up the air in the cup holder, the table may be attached to any cup holder and so remain steady. As shown in figure 13. Wood is also used for support.

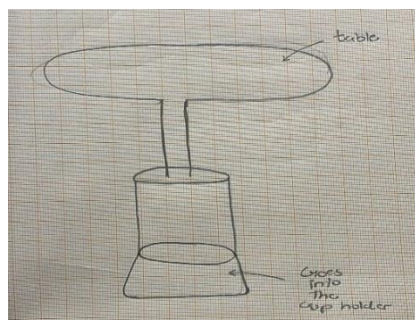


Figure 13: Concept B

Concept C)

The table has a 0.7m² surface area and is composed of plastic-coated leather for a comfortable feel and attractive appearance. The weight is estimated to be 0.4kg. By sucking up the air in the cup holder, the table may be attached to any cup holder and so remain steady. It has a joint that allows movement along the horizontal path, allowing for increased movement flexibility. The support is made of aluminum, which is lightweight but durable enough to sustain the weight.

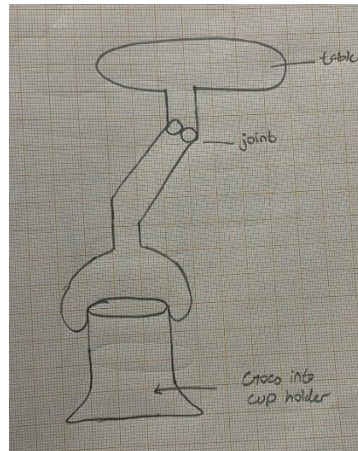


Figure 14: Concept C

Concept D)

The Concept D table features a unique design that allows it to fold twice, allowing for more dining room. It has a 1m² surface area and uses the same attachment principle as concepts B and C. It's comprised of light plastic with a luxurious-looking leather finish. The weight is estimated to be 1.5kg. It also features supports at the ends of each edge that may be attached for added stability. Furthermore, because it has four legs that can be attached to the borders of each corner, this table can be utilized as a portable table. As shown in figure 15.

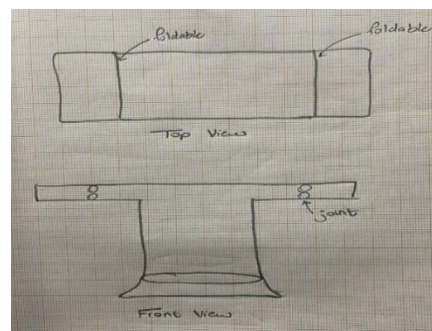


Figure 15: Concept D

Concept E)

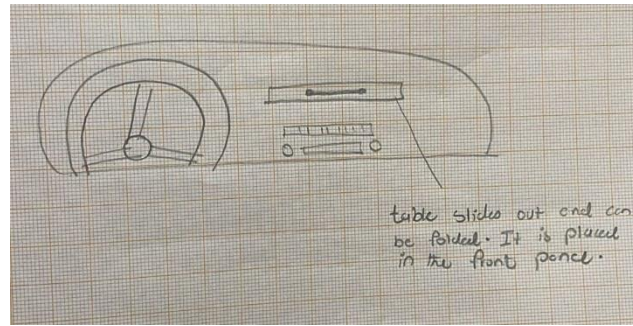


Figure 16: Concept E

The table is mounted to the front panel of the car under the glass in this concept. It can slide in the direction of the passengers and then fold, allowing it to reach the end of the front seats. The support's one face is then secured to the table's end, while the other is secured to the gap between the two front-seat occupants. It's made of lightweight plastic that's also tough enough to resist bending. As shown in figure 16.

7.0 Prototyping

Prototyping is an important part of the design process as it assures that the product's components and subsystems behave together as intended. It will show that the product has obtained a certain degree of functionality. The team wanted to construct a physical prototype however, due to time and financial limitations only a rough analytical prototype was constructed using Solidworks as shown in figure 17 and figure 18.

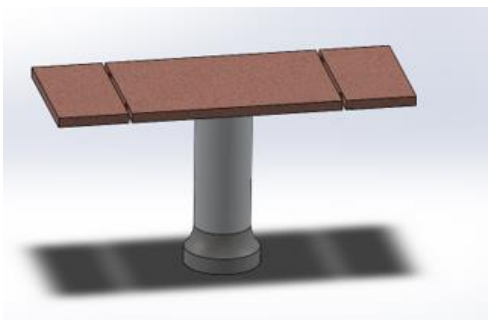


Figure 17: Prototype 3d view

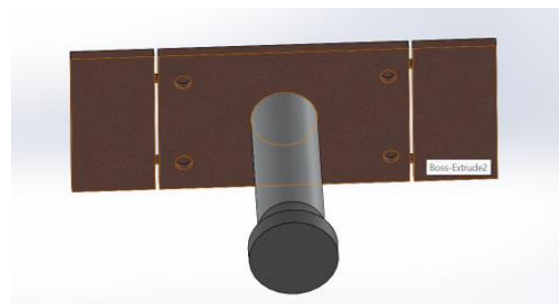


Figure 18: Prototype bottom view

8.0 Concept Selection

After identifying the criteria, the group used the (0, +,-) scale to compare and rank the concepts. concept B was chosen to be the benchmark reference concept because it is a straightforward concept with which the team members are familiar. The other concepts were compared with it, then after the scaling was done, the concepts were ranked as shown in table 4. With some improvements made, the group decided to take concepts C, D, and E to the next stage, the scoring stage.

Table 4: Concept-Screening matrix

	A	B (Reference)	C	D	E
Cheap sell price	+	0	-	-	0
Light Weight.	+	0	+	-	0
Large surface area	-	0	0	+	+
smooth surface	-	0	+	+	+
good grip surface	-	0	+	+	-
Assemble/disassemble time	-	0	0	+	+
Durability	+	0	0	+	-
sum of +	3	0	3	5	3
sum of 0	0	7	3	1	2
sum of -	4	0	1	2	2
rank	5	4	2	1	3

After that, the group used concept C as a reference and proceeded with the concept scoring matrix as shown in table 5. In the end, the group decided to continue with the best possible concept to develop, which was concept D since it was ranked the highest.

Table 5: Concept-scoring matrix

		C		D		E	
	Weight	Rating	Weighted	Rating	Weighted score	Rating	Weighted score
Cheap sell price	20%	3	0.6	3	0.6	3	0.6
Light Weight.	10%	2	0.2	2	0.2	4	0.4
The large surface area	30%	2	0.6	5	1.5	4	1.2
smooth surface	5%	3	0.15	4	0.2	3	0.15
good grip surface	10%	3	0.3	4	0.4	2	0.2
Assemble/disassemble time	10%	3	0.3	4	0.4	2	0.2
Durability	15%	3	0.45	2	0.3	1	0.15
Total score		2.15		3.6		2.9	
rank		3		1		2	

9.0 Concept Testing

The study of prospective consumers' responses to a product idea before it is launched in the market is defined as concept testing. As firms and organizations prepare to launch a product, concept testing is a crucial step in determining impressions and needs. It can assist in determining and enhancing the product's target market. The goal of testing is to discover whether the product will be successful or not, as well as to choose the best market. Everyone who owns a car, as well as hikers who seek portable furniture, are included in the survey population. To evaluate the concept, the survey format was conducted. Figure 19 and figure 20 display a sample of the questions and the responses received from the customers.

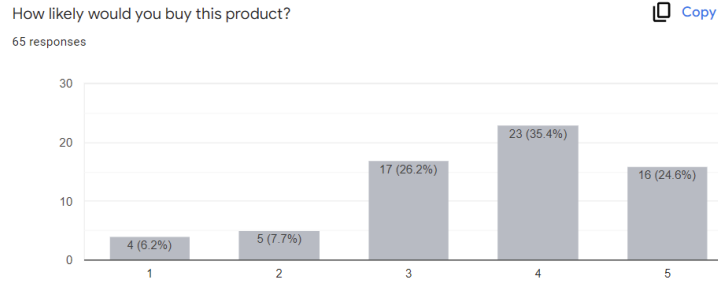


Figure 19: Survey question sample

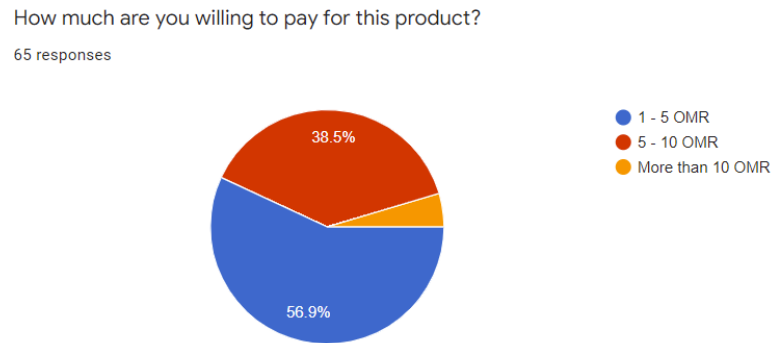


Figure 20: Survey question sample

10.0 Design for Manufacturing

Design for production was considered throughout the development phase of this project. One of the team's first objectives was to have as few parts as feasible. As a result, most designs do not require screws or bolts for assembly or disassembly after production. Second, because the chosen concept's general design is made up of basic shapes, the manufacturing procedure will be simple. In addition, when choosing the table material, the team considered the simplicity of manufacture as well as the relative costs of producing the product with that material. As a result, for all concepts, plastic, wood, and aluminum were picked as alternatives. Furthermore, due to a lack of information, it was difficult to predict how the pieces would be assembled after manufacture. However, after doing a study, the cheapest and most appropriate assembly operations for the offered concepts are mechanical and welding assembly.

11.0 Managing Projects

Without planning and management, almost nothing can be accomplished. The development team can use project planning to coordinate the stages of product development. So, first, the team gathered all of the project's activities, and then, as shown in figure 21, a PERT chart was created to highlight the precedence connection and the overall time required to complete the project. In addition, as illustrated in figure 22, a Gantt chart was created. Finally, as shown in figure 23, another Gantt chart was created to show how things may have gone differently if the team had done the project again.

Activities:

1. Looking for opportunities to pursue.
2. Choosing the best opportunity to go with.
3. Gathering customer statements from one survey and interviews.
4. Change customer statements to customer need statements.
5. Generate target specifications.
6. Patent search
7. Generate concepts.
8. Select the best concept.
9. Design for manufacturing
10. Management of the project
11. Product development Economics
12. Final report

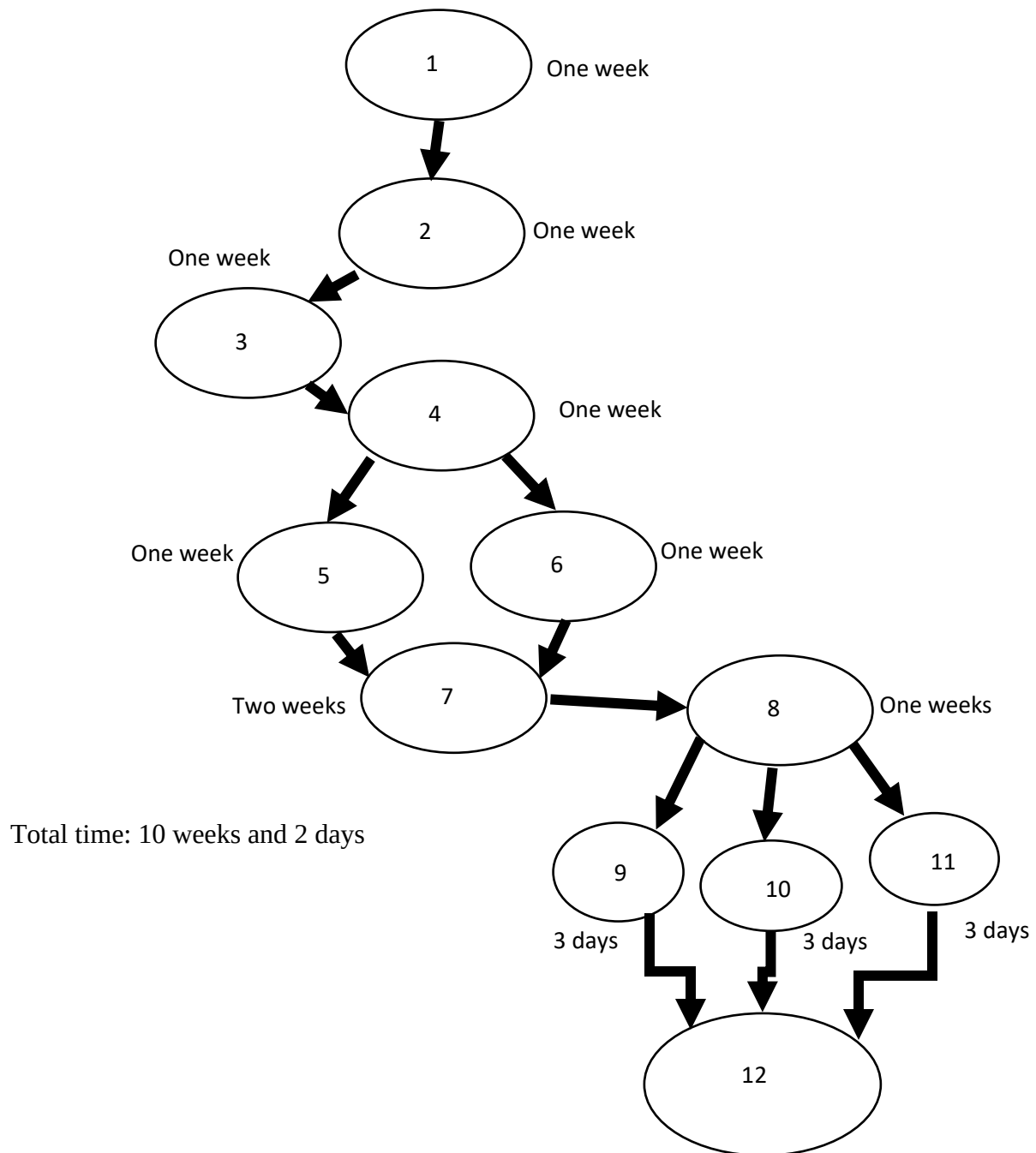


Figure 21: PERT chart

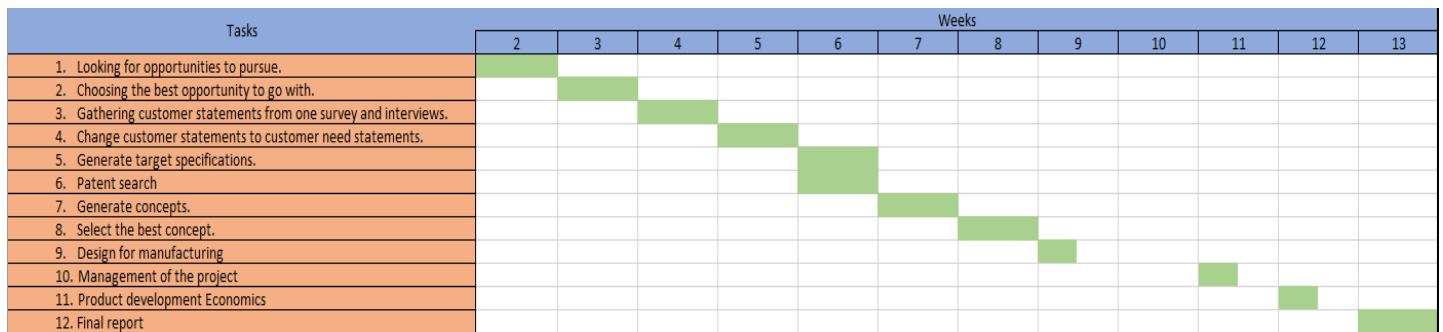


Figure 22: Project Gantt chart

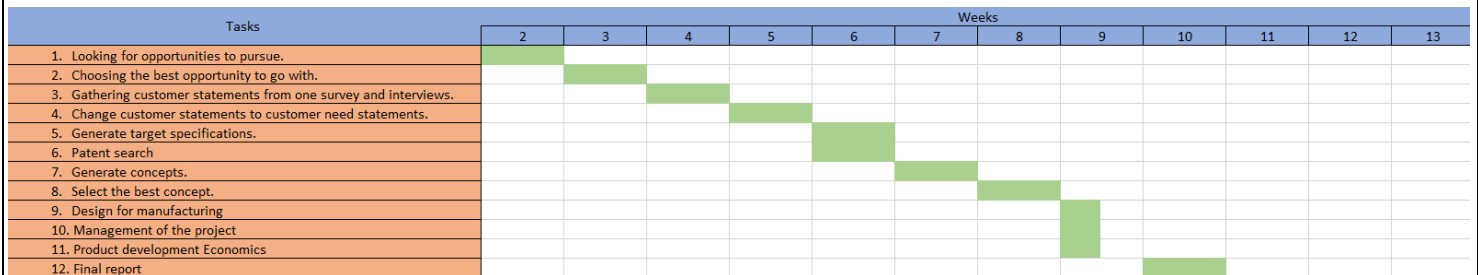


Figure 23: Improved Gantt Chart

12.0 Product development Economics

To make sure that the product is economically justified, the team constructed an economic analysis using a file provided by the Dr. The discount rate was chosen to be 1.75%, and the approximate project lifetime is planned to be 10 years. The results of the analysis and further details are shown in the diagrams and table below.

Table 6: Inputs to the economic model

	first	last	burn rate
Development	1	2	-4000
Testing	2	3	-2000
Tooling and Ramp-Up Costs	3	4	-12500
Market Introduction	3	4	-3000
Ongoing Marketing Costs	3	10	-2000
Unit Sales	3	10	2000
Unit Price	3	10	7.000
Unit Production Cost	3	10	3.000
Discount Rate (per time period)		1.75%	

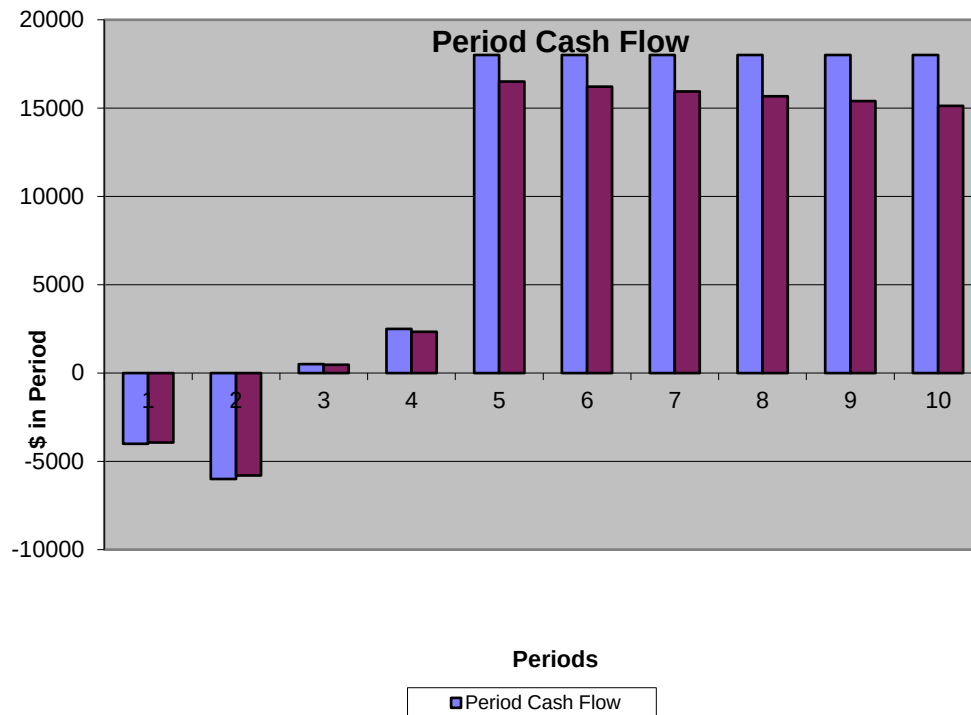


Figure 24: Period Cash flow

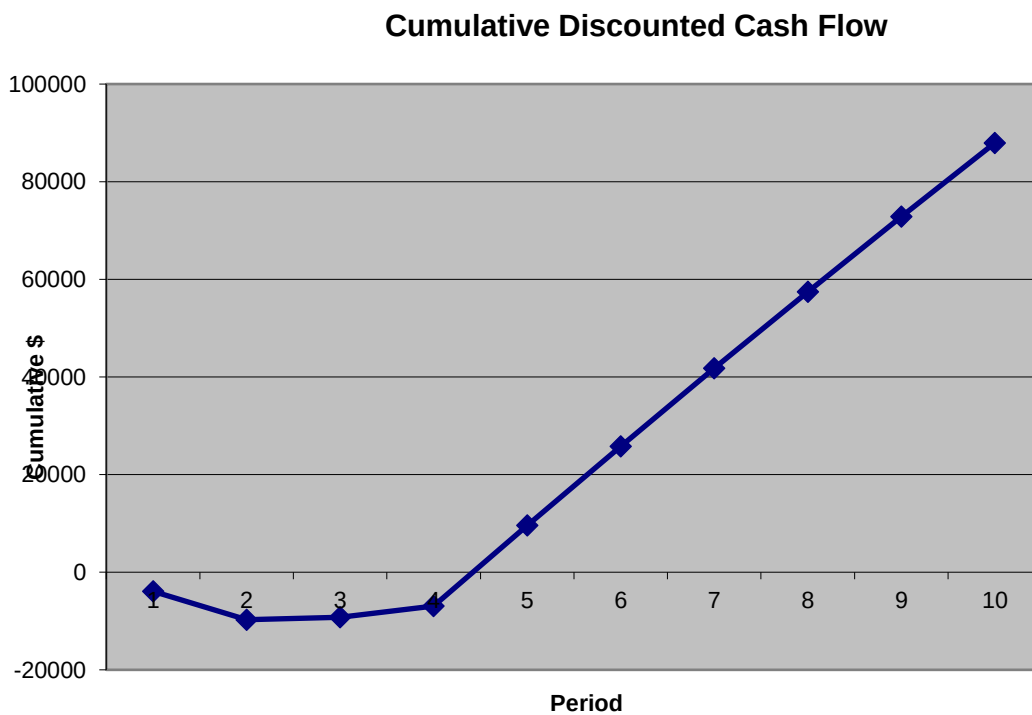


Figure 24: Cumulative Cash flow

13.0 Conclusion

The product development process involves transforming concepts into innovative and successful products to meet unfulfilled requirements or improve an existing solution. The development process was used to create a new product in this project. The goal was to give customers a portable car table that would meet their needs. The team's biggest challenge in the project was the lack of technical understanding. The team hopes to create a real prototype in the future to test and verify if the product fulfills market demands.

14.0 Reference

- [New Cars, Car Reviews, Car Shows, Car Photos - The Car Connection](#)